

Final Draft

Exploring Sensor-Based Wearable Technology For Ballet Training

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Abstract:

Ballet training imposes high economic and personal costs to dancers with a recorded dance injury rate of approximately 1-4 injuries per dancer per 1000 dance hours across the career of a dancer. It warrants the development of purpose/supervision-built training shoes distinct from performance footwear. Informed by this context, this paper proposes a human centered design intervention towards developing purpose built training shoes in addition to the existing performance shoes available in the market. Towards this goal this paper provides a systematic review of challenges of being a Ballet dancer focusing on a) Biomechanics and Training related demands for Ballet dancers b) Common Injuries in Ballet Training c) Costs and Risks associated with training. Further, it outlines the proof of concept of a functional shoe prototype “Equipointe” as a productable solution for the challenges. This Paper describes the working of this Ballet Dance Training Device that works on the principles of “Human Centered Design” as a Pedagogical Tool for Teaching Ballet. Towards conclusion it provides an overview of cost and demand associated with product development and provides potential market assessments for the same.

Keywords:

Wearable technology, performing arts education, ballet, sensory data, body movement, gait analysis, body risk analysis, cost of prototype, material, supply chain, market entry strategy

1. Introduction

Ballet is considered one of the most physically demanding dance forms in the world, requiring a range of movements and intense technique. It requires both immense flexibility and physical strength to achieve an optimal range of motion and aesthetic presentation and to be able carry out movements and hold positions on stage. Strength is specifically prioritised in the feet, core, ankles and legs. Strength in the core is needed for postural stability, the feet for articulation and jumps, the legs for jumps and extensions and the ankles for pointe work. Furthermore, ballet requires balance, especially during movements like turns and slow variations. It may also be needed for something as simple as holding a position for a prolonged period of time. Additionally, one of the most essential parts of ballet - grace - relies heavily on coordination. The ability for the arms and legs to move in sync with one another enhances the aesthetic of the movements and creates a feeling of elegance and grace, two defining traits of ballet. While these are few of the main physical

demands that the dance requires, the list goes on. It is worth noting, however, that each of these physical demands are heavily interdependent on one another. For example, with strength comes balance, with balance comes coordination and a combination of strength and flexibility are needed to achieve technically sound movements. Thus overall, ballet represents a complex form of art, heavily reliant on numerous interdependent physical factors that contribute to the overall feel and nature of the art.

Ballet started as a form of elitist entertainment performed in Russian Courts in the 15th Century. Ballet was performed not just as a form of entertainment, but also as a demonstration of power and prestige. Individuals used ballet as a means of displaying their wealth, demonstrating cultural superiority and reinforcing royal power. Over time, it expanded to different countries in Europe, namely Italy and France, and 2 distinct ballet forms emerged through this transition: vagonova, French-Italian. Vagonova, originating in Russia, focuses more on flexibility and perfect technique whereas the French-Italian approach employs more elegant and free movements for the dancer (

Becoming a full-time ballet dancer, working for a company requires extremely intense training. Physically, one must dedicate about three to four hours a day, five to six days a week, including plenty of pointe work. In addition, dancers should be flexible and versatile in terms of styles, so complementing their ballet technique with a corrective tool would be helpful. Although the dance is inherently challenging, wearable technology is not widespread. In the past, ballet relied upon aesthetics, mirrors, and personal observations of instructors. Contrary to popular sports that embraced such tools as GPS trackers, IMUs and biometric sensors, ballet has been late in adopting wearable technologies because of its delicate and sophisticated nature. Nevertheless, it has been shown that wearable technologies may have a considerable effect on ballet dancers' skills. Wearable technologies do not protect dancers from injury but they give the necessary data that will help develop an evidence-based strategy for injury prevention. For instance, studies show how 82% of injury rates among professional ballet dancers have been reduced with the help of the Injury Prevention Program (IPP) that has been developed by specialists of Houston Ballet and Houston Methodist. However, wearable technologies can be used to implement similar programs for injury prevention. As ballet continues to embrace innovation, technology is becoming an increasingly valuable tool that complements, rather than replaces, expert coaching, allowing dancers to train more efficiently and sustainably. (*An Injury Prevention Program for Professional Ballet: A Randomized Controlled Investigation by Houston Methodist*, 2020)

2. Literature Review

2.1 Biomechanics and Injury

Ballet biomechanics involve extreme precision, manipulation of human anatomy, utilising external hip rotation, vertical alignment and high velocity movements that often produce ground reaction

forces 4-5 times a dancers' body weight. It centres around optimising energy transfer and stability through controlled muscle torque and various rotational mechanisms.



AI generated Image

Ballet biomechanics runs on two primary principals: Newton's laws of motion in ballet and balance and stability. This reflects the fact that movements in dancing are deeply scientific, rather than arbitrary. Newton's first law related to inertia can be seen when a dancer rotates in a pirouette, for example, as they continue rotating until obstructed by an opposing force, like friction or muscle breaking. Newton's second law, $F = ma$ is seen through jumps, where jump height is dependent on the force exerted on the floor relative to body mass. Lastly, Newton's third law on action and reaction forces is reflected through jumps as well as the downward push of the foot on the floor generates an equal and opposite force that propels the dancer into a jump. Furthermore, stability and balance in ballet depend on three key factors. First, the centre of gravity. Typically located near the pelvis, the centre of gravity must be aligned with the supporting leg to ensure balance while dancing. Second, the base of support (the dancer's feet or single foot) requires neuromuscular control. Third, alignment plays an important role as the shoulders, knees and hips need to be 'stacked' on top of each other to minimise compensatory limb activity and maximise efficiency.

Research identifies four biomechanically grounded pillars of classical ballet technique: turnout, the external rotation of the hips; alignment, the vertical stacking of joints to minimise shear forces; placement, the optimal positioning of body segments relative to the centre of gravity; and extension, the controlled elongation of the limbs while maintaining alignment.

The activation of lower body muscles and joint kinetics in ballet dancing rely on the interplay of the quadriceps, hamstrings, glutes, calves, hip flexors, and deep external rotators. This helps with the creation and control of motion at the hip, knee, and ankle joints. The role of the core and spinal muscles is to stabilize the trunk and facilitate the efficient transfer of forces. The role of the quadriceps is to extend the knee joint during jumps and control landing. The role of the hamstrings and glutes is to facilitate hip extension in moves such as arabesque and stabilization of the pelvis. Calves create plantar flexion for relevé and pointe work and cushion the impact during landings. Hip flexors are involved in the elevation of legs in moves such as développés and grands battements. The deep external rotators of the hips help create turnout of the legs without compensation at the knee and ankle joints. In the upper body, scapular stabilizers and shoulder musculature support port de bras and partnering, integrating with lower-limb kinetics so that ground reaction forces, joint torques, and muscle activation patterns combine to create both the aesthetic lines and the mechanical efficiency characteristic of classical ballet technique.

Due to the physically demanding nature of ballet as an activity, dancers are prone to numerous injuries. Ballet causes achilles tendonitis due to repetitive, high intensity strain on the tendon, biomechanically known as ‘overuse.’ It may also be caused due to excessive **plantarflexion**¹. Early symptoms include heel/ankle pain, stiffness and swelling. Ankle sprains may also be caused due to the errors in technique ‘sickling’ and ‘rolling.’ Sickling is when the alignment of the foot gets disrupted, while rolling is when the weight is distributed unevenly on the sole of the foot. Early symptoms of this include sudden pain, swelling and reduced mobility. Stress fractures are another such injury caused due to poor weight distribution on the sole of the foot. Early symptoms of stress fractures include persistent localised pain and swelling. Plantar Fasciitis is caused due to repetitive jumping movements (known as allégro in ballet), repetitive relevés (rising) and working in pointe shoes which leads to micro tears in the fascia. These injuries are augmented by dancing on hard, non conducive surfaces and maintaining a constant, high tension stretch in the plantar fascia during intense training. Ballet causes patellofemoral pain primarily through repetitive overloading of the knee joint, often stemming from poor alignment, weak glute muscles and forced turnout. Lastly, spondylolysis is a stress fracture in the lower back caused primarily through repetitive hyperextension (arching) of the lumbar region caused by certain movements in ballet.

2.2 Injury burden and prevalence

¹ Plantarflexion is the movement of the ankle where the foot points downward, away from the shin, increasing the angle between the foot and leg.

Injuries among professional and pre-professional ballet dancers occur at a similar rate as for elite athletes in high-impact sports but with a characteristic occurrence of overuse injuries instead of acute injuries. Prospective monitoring data from professional ballet companies indicate 4.4 injuries per 1,000 hours of dancing, with an average of 6–7 injuries per year for each dancer; over several years, the incidence proportion is between 47–73% per year. It translates into almost 50–75% of dancers who will be injured at least once during the season and many dancers with multiple episodes. Around 80% of dancers suffer from at least one injury annually, while around 16% have suffered from three or more injuries within the year.

However, the majority of these injuries have to do with overuse, which accounts for between 50 to 68 percent of all reported injuries, while the rest are considered to be traumatic. The predominance of these injuries can be explained by the nature of ballet, where dancers are subjected to repetitive, high-volume training that includes numerous jumps, relevés, and range-of-motion activities. In contrast to contact sports injuries, which are frequently caused by collisions, in ballet, the injuries occur due to the accumulated microscopic injuries in the tendons, bones, and joints.

Recurrence is a common aspect of the burden of injury in ballet dancing. Many dancers keep on training despite their injuries due to the schedule for performances and fear of losing their jobs or roles. In the process of "dancing through pain," it becomes hard for dancers to recover and they keep getting injured again. Statistics show that out of all the injuries causing loss of training days, 35% of the injuries need more than 28 days of modified training. However, many dancers continue training before fully recovering. It means that injuries keep occurring in the same location over and over again.

In terms of its injury prevalence per 1,000 hours of activity, ballet can be comparable or higher than elite soccer, basketball, and gymnastics when looking at injuries of the lower extremities associated with overuse. However, the issue is not sufficiently researched and lacks standardization within the field. This is why the actual prevalence might be lower than reported. The problem of injuries is important not only due to its negative effects on dancer's health and ability to perform at his best but also because it represents a cost in terms of economics and artistic performances.

2.3 Performance costs

Injuries in ballet take a steep toll on performance, extending far beyond the immediate pain or tissue damage. The consequences manifest as measurable declines in technical execution, physical capacity, and career trajectory, with compounding effects across seasons.

2.3.1 Reduced technical execution and artistic quality

Though dancers may dance even if injured, compensation through biomechanics is bound to affect the performance negatively. Pain and neuromuscular changes affect precision of turnout, extension, and alignment, which are core concepts in ballet. Dancers suffering from ankle or foot injuries show lower plantarflexion, thus affecting the ability to perform relevé and pointes. Injuries in the lower limbs affect the generation and absorption of force, thus affecting the height of jumps

and landings, and in fact, one particular study showed that dancers who were injured were able to generate lower ground reaction forces during sauté, and this affected their jumping ability. Deficits in balance become evident as instability of the ankle joint and hip joint interfere with the ability to make postural adjustments needed during arabesque and adagio balance poses.

2.3.2 Decreased endurance and movement efficiency

An injured dancer consumes more energy for the identical motions as he/she relies on compensatory muscle usage and joint mechanics that differ from the healthy case. Studies indicate that when dancers practice under pain, they experience faster onset of fatigue and decreased aerobic capacity while attending classes and rehearsals. Overuse injuries account for 50-68% of all ballet injuries and are very deceptive since they slowly decrease the level of stamina over weeks or months without causing an immediate withdrawal of the dancer. Dancers who suffer from tendinopathy or stress reactions experience a 20-30% decline in stamina and are forced to make changes during rehearsals or withdraw prematurely.

2.3.3 Missed rehearsals and performances

Data on time losses show the immediate effect of injuries on performance, which is 7 days of reduced or no training on the average for professional ballet dancers, while on the average male ballet dancers have 9-10 days of such losses per injury. Severe injuries, especially those involving the foot, ankle and knee joint, lead to 28 or more days of restricted activity in 35% of instances. During the regular season, this amounts to 48-60 days lost by each dancer for full training and performance due to injury. Principal dancers have even greater risks as they sustain 2.0-2.2 more injuries requiring medical attention per 1,000 hours of dancing than do apprentices, which means that more demanding performers are subjected to greater disruptions.

2.3.4 Interrupted skill acquisition and delayed career progression

Injuries for pre-professional and beginning professional dancers disrupt the vital period of skill acquisition. The learning process involved in ballet dancing requires repetitive and quality practice sessions; interruptions in this regard affect motor learning and neural adaptation processes. Dancers taking off 2–4 weeks during the periods of intense learning lag behind other dancers in skill acquisition of difficult exercises such as multiple pirouettes, grand allegro, and partner lifts. This lagging is significant in terms of future professional progress, because less experienced dancers have significantly longer absence from dance after an injury than their more experienced colleagues, thus developing slowly in terms of technical advancement. In case of repeated injuries, over the course of their professional lives dancers are not able to get promoted to be a soloist or principal dancer because they are viewed as unreliable or physically unprepared. Historical data show that dancers having three or more injuries annually have less chances to become a principal dancer within 5 years.

2.4 Economic Costs

However, the cost of injuries to a ballet dancer's career is not confined to the expenses directly incurred due to injury, but also involves indirect costs from lost performances, along with long-term economic ramifications from a career perspective. An extensive review of workers' compensation claims among a large professional ballet troupe for three consecutive seasons

showed that 104 dancers suffered 309 injuries that involved total medical expenses of \$398,396, which translated into \$1,289 per injury. It was noted that while only 4.2% of injuries were of more than \$5,000 medical costs, they made up 60% of the total amount spent, and there were nine injuries involving over \$10,000 each.

2.4.1 Direct medical rehabilitation costs

The direct costs of treatment of ballet injuries consist of physiotherapy appointments, sports medicine consultations, medical diagnostic imaging, and rehabilitations. Specialized physiotherapy sessions cost between \$90 and \$135 on a regular basis and between \$135 and \$195 for the pointe readiness screening appointment and written report. The average number of rehabilitation sessions required to treat an overuse injury is 8 to 12 sessions in 6 to 8 weeks, which amounts to \$720-\$1620 only for the physiotherapy itself. The consultation with sports medicine doctors that might be needed in case of difficult or long-lasting injury adds \$200 to \$400 to the costs, and the MRI or X-ray imaging for the foot, ankle, and knee injuries add another \$400 to \$1500 per diagnostic procedure. The strength and conditioning exercises to avoid injuries cost \$100-\$200 per month.

2.4.2 Indirect costs: equipment and training modifications

Injuries sometimes lead to equipment replacement and changes in training programs. The use of pointe shoes is doubled by dancers who have altered their load distribution patterns because of foot and ankle injuries and must therefore replace their shoes every 2–3 weeks as opposed to 4–6 weeks. This costs an additional \$1,000–\$2,000 since pointe shoes cost \$80-\$120 each. Moreover, the dancers might have to buy orthotic devices, braces, or compression clothes that can cost from \$100 to \$300.

2.4.3 Lost income from missed performances

The most critical economic consequences come from the loss of professional earnings. In companies not having guaranteed salaries, typical of the modern small ballet companies, the dancers earn by the performance or have a lower salary while on sick leave because of an injury. Thus, the loss of earnings for a principal dancer, being off work due to injury for 4 weeks, will be about \$3,000-\$8,000, depending on the pay system in the company. For soloists and corps de ballet, it would be less but still significant: \$1,500-\$4,000 per month of absence from the stage. Taking into account the fact that the average number of days of injury is 7-10 per injury and that each dancer has 6-7 injuries per season, the total loss of earnings in the career of a dancer is significant. The dancers who experience injuries repeatedly, 24 dancers were found to suffer from 5 or more injuries each, which accounts for 51.9% of all injuries, have the largest financial risks.

2.4.4 Long term career and retirement costs

However, the economic ramifications go further than just the period of injury. Dancers who suffer several major injuries will retire early due to their injuries, reducing their total potential earning capacity by five to ten years. The typical professional ballet dancer will work for 15 to 20 years; retiring at the age of 28 due to an injury rather than 35 means losing \$150,000-\$300,000 in total earning power, depending on the level of company and dancing part. Once they leave their careers,

those dancers with long-term injuries must continue to pay for treatments such as pain management, joint replacement surgeries, or physical therapy up until their 40s or 50s.

2.5 Limitations of current prevention

Injury prevention measures employed today in ballet dancing consist of observation, periodic and reactive methods, including teacher observation in class and rehearsal, physiotherapist screening at the start of seasons or post-injury, strength and conditioning training carried out in an intermittent fashion, and periodic biomechanical analysis done either in laboratory or clinical setting. Although all these measures have proved their worth, structured injury prevention programs capable of lowering the injury incidence rate by up to 82% when continuously implemented, are nonetheless plagued with certain inherent limitations.

2.5.1 Teacher Observation

Teachers and ballet masters act as the primary form of defense, with their professional eye for recognizing technical flaws that increase susceptibility to injury. This is a highly subjective process that depends entirely on the individual's experience and skills, however. Even one teacher managing a class of 20–30 students cannot keep track of all aspects of the dancer's alignment, strength and joint mechanics while performing various exercises. Small changes in the dancer's technique, like losing 5 degrees of turnout from the hip, developing knee valgus in plié, or uneven distribution of weight in relevé, will not be noticed by just looking at the dancer. Additionally, observation is an intermittent process: it takes place only during classes and rehearsals, and provides no information about how the dancer practices or performs outside of those classes.

2.5.2 Physiotherapist Screening

The objective of physiotherapy screenings done during pre-season training or after injuries is to detect strength imbalances, mobility deficits, and movement dysfunction before they turn into injuries. For instance, all dancers who participate in the Australian Ballet's programme undergo screening to determine calf endurance, hip mobility, and foot muscles strength and are given appropriate exercises. Despite being very useful, screenings give only a snapshot of the dancer's condition at a particular point in time. For instance, the dancer can easily have enough of calf endurance in September but will experience deficits because of fatigue by November because of heavy rehearsals and be detected only after showing any signs of symptoms. Furthermore, physiotherapy screenings consume a lot of resources such as time, skilled workers, and sometimes expensive equipment like force plates and motion capture.

2.5.3 Strength and conditioning programmes

Conditioning programs that are evidence-based, such as the 11+ Dance neuromuscular warm-up program and the company-wide calf endurance programs, have been proven to be effective in minimizing the injuries of the lower limbs. Nevertheless, the said programs have usually been prescribed to all the dancers without taking into account real-time biomechanical information. A dancer who has a hidden ankle instability would be subjected to the same relevé program as another dancer who has hyper-mobile ankles, yet both dancers have completely different risk for sustaining an injury. In addition, compliance to these programs remains poor. Dancers tend to forego conditioning exercises during high-intensity rehearsals.

2.5.4 Periodic biomechanical assessments

The analysis of kinematics, muscles activity, and kinetics in laboratory settings through the use of motion capture, force plate, and EMG offers great insight into the body mechanics and muscular action of an athlete or performer. But these evaluations are done only sparingly, usually only once or twice a season, in controlled settings that do not match the unpredictability of a real performance situation. A dancer could have perfect technique in a laboratory setting but compensate for the fatigue and stress of an entire production by falling differently in a live performance.

2.5.5 General systematic limitations

The ultimate limitation of the existing prevention efforts is their reactive approach in that the interventions are launched only when there is an obvious symptom of overload or an actual break in a technical routine. The overload, which is the main cause of overuse injuries accounting for 50-68% of all ballet injuries, often remains unobserved until there is already a damage done to tissues. It may take a couple of weeks for dancers to make thousands of jumps, relevés, and other movements requiring extreme ROM without any measures being taken to assess their cumulative overload.

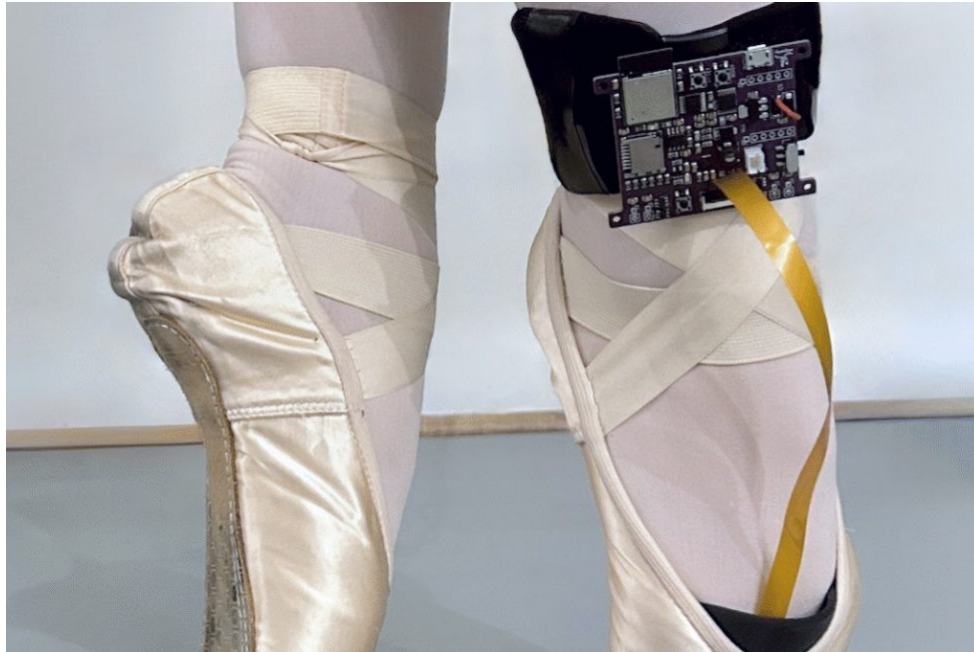
The aforementioned responsive model is further aggravated by a "dance through pain" mentality whereby dancers hide their symptoms in order not to be sidelined or viewed as unfit for the job. In absence of constant measurements of load, quality of techniques, and level of stress, teachers and healthcare professionals cannot act before the development of an injury. The outcome is a preventive framework which reacts to issues at the wrong time, treats only its symptoms, and does not take advantage of the window of opportunity during which relatively small changes can help avoid serious injuries.

One of the consequences of such challenges has been the interest shown by researchers in using wearable technology as a means of continuous measuring of biomechanical parameters.

3. Wearable technology in the ballet industry

Numerous studies and literature on wearable technology in the ballet industry have been analysed to assess its scope in the ballet world, its limitations and current adoption.

Research into a wearable ballet device utilized 64 flexible pressure sensors embedded in a toe-pad insert to analyze plantar pressure distribution, providing real-time haptic feedback via an ankle band. The study highlighted significant design challenges: the necessity for diverse datasets to account for varied foot anatomy, the difficulty of standardizing a product tailored to individual dancer needs, and the engineering hurdle of maintaining a lightweight, unobtrusive, and untethered design within the restrictive space of a pointe shoe (*Engineering Meets Ballet: A Smart Solution for Safer Pointe Work*, 2025).



Another study was a comparison between new pointe shoes that had 5-50 hours of training and dead pointe shoes that had 51-400 hours. The movements tested on dancers were relevé à la second, sous-sus and a single pirouette. The measuring the centre of pressure sway area using a force plate and Ground Reaction Forces (GRF) in three directions: anterior-posterior, medial lateral and vertical the following was found: The centre of pressure sway area, using a force plate and Ground Reaction Forces (GRF) was measured in three directions: anterior-posterior, medial lateral and vertical. The findings were that “Dead” shoes showed a significantly larger ‘sway area’ from the mean centre of pressure, impacting stability; No statistically significant differences in the force directions, vertical, medial and lateral, were found across the two shoe conditions; and “dead” shoes provided less stability and reduced balance, whereas ‘new’ shoes allowed more force production in certain movements (Academy, 2019).

A study conducted on testing a wearable insole system to measure plantar pressure and shear stress for diabetic foot ulcer prevention details the design, development and testing of this system. The system utilises thin, flexible TRIPS sensors placed at the heel, 1st and 5th metatarsals and the hallux. The sensors recorded pressure and shear in two orthogonal directions and are integrated into an insole. The final product showed high accuracy demonstrating that the multi-sensor approach and flexible insole design is feasible for monitoring pressure distribution in pointe shoes (Tang et al., 2023).

Another project aiming to reduce high pressure on a dancers’ foot in pointe shoes, by specifically improving upon the pressure distribution in the toe box of the pointe shoe focused on different materials inserted into the box, to investigate which was most effective in redistributing pressure amongst the toes with a specific focus on reducing pressure on the big toe. The materials tested

include spray foam, soft foam, rigid plastic, memory foam and more. Pressure indicating film was used to identify the pressure distribution and it was concluded from the experiment that amongst all materials tested, spray foam performed the best as it had the lowest average percentage area of the toe that was unloaded; the second best was soft foam and the worst material was rigid plastic; and spray foam was also liked by consumers as they found it improved support and comfort ([The Pressure Pointe - Sensorprod, 2023](#)).

Furthermore, a study was conducted on the most effective sensing technologies used to assess motor behaviour on ballet. After examining various studies, it was concluded that force plates, electromyography (EMG) and inertial sensors are the most frequently used methods for evaluating classical ballet movements ([Quadrado et al., 2022](#)).

Moreover, an investigation into a novel pressure measurement system, leveraging thin, flexible capacitive sensors embedded in pointe shoes to quantify toe pressure distribution during ballet movements. The sensors, pulled across the toe box, captured real-time data on pressure hotspots in professional dancers performing relevés and jumps, revealing peak loads exceeding 200% body weight on the first metatarsophalangeal joint. Key findings include significant pressure asymmetry between toes, with the hallux bearing 40-50% of the total load, and correlations between sensor data and self reported foot pain ([Miura et al., 2025](#)).

A paper evaluating a shoe-mounted accelerometer and gyroscope system for assessing ballet jump height, landing forces, and aerial time in variations like sautés and jetés was worn by 12 dancers. The IMU sensors provided 95% accuracy against force plates, identifying technique flaws linked to sesamoiditis from harsh landings. The untethered setup enabled field testing without lab constraints. Another key finding was that real time feedback improved landing symmetry for jumps ([Almonroeder et al., 2019](#)).

Another research investigation used force-sensitive insoles in pointe shoes to analyse pressure patterns across 20 professional ballerinas during plies, relevés and pointe work. Results showed metatarsal heads endured 1.5-2.5x normal loads, with rigid shanks exacerbating hallux rigidus risks; flexible toe pads mitigated peaks by 25%. Two challenges with respect to the investigation were revealed: individual foot morphologies requiring custom sensor arrays and long term wear causing sensor drift ([Pearson & Whitaker, 2012](#)).

Researchers developed a wearable insole with 16 FSR sensors integrated into pointe shoes to map plantar pressure during dynamic ballet tasks like pirouettes and grand battements. The system logged data at 100 Hz, showing "dead" shoes increased peak pressures by 15-20% under the metatarsals compared to new ones, heightening injury risk. Haptic feedback prototypes reduced overload incidents by 30% in trials with elite dancers. The study revealed three primary notable outcomes: there was a higher sway in centre of pressure with shoes that were "dead," there is a

need for dancer specific calibration, and a lightweight design under 20g per shoe is an ideal architecture (Sato et al., 2025).

4. Designing a Ballet Dancing Show

4.1 System design



4.1.1 Hardware overview

The Equipointe prototype combines pressure sensing, inertial motion detection and wireless data transfer capabilities within the design of a regular ballet pointe shoe to facilitate biomechanical analysis during dancing. The components include an ESP32 controller, four Force Sensitive Resistor (FSR402) sensors, an MPU6050 IMU, a rechargeable LiPo battery and a wireless module. The components are externally attached to the shoe to ensure minimal obstruction of the dancer's performance yet to permit easy prototyping and further hardware development.

The FSR402 sensors are installed in the areas of the shoe that experience significant loading forces during ballet, such as the toes, forefoot, and heel, to analyze changes in ground reaction forces when performing dance moves. The sensor readings are obtained through the ESP32's analogue to digital converter, where the sampling is performed on each channel at a constant sampling frequency.

Rotational and translational movement is detected by an MPU6050 Inertial Measurement Unit (IMU), which uses three-axis accelerometer and gyroscope. The readings from the sensor are communicated via the I²C protocol to the ESP32 and used to obtain information about linear

acceleration and angular velocity of the body. The results are further utilized for orientation estimation through the Madgwick sensor fusion algorithm.

The ESP32 acts as the CPU to coordinate all the activities of sensor data acquisition, recording timestamps of these data, and wireless transmission of data to the Python-based dashboard for visualization and data storage. This dashboard will show the pressure and motion data in real-time while at the same time storing the data in CSV format. Figure X shows the prototype and integration of all the hardware components used in the system.

4.2 Sensor integration and principal of operation

4.2.1 Force Measurement

Four FSR402 sensors are distributed throughout the sole of the pointe shoe to detect changes in the loading of the foot. These sensing points reflect areas that are expected to undergo different loading patterns while performing ballet steps: first metatarsal, fifth metatarsal, heel, and anterior/toe areas. The use of several sensors instead of one pressure measuring point allows for the detection of changes in loading distribution within the foot.

FSR stands for Force Sensitive Resistors and, as the name suggests, have an electrical resistance that reduces with an increase in the applied force. These devices are integrated in voltage dividers, such that change in resistance translates to change in voltage readings. The ESP32 reads these analogue voltage measurements using the analogue-to-digital converter. This means that the obtained signal is dependent on time.

The value obtained from FSR is not used directly as a measure of the force. Instead, each of the FSRs needs to be calibrated for the applied force. It should be noted that calibration is necessary for FSR sensors due to the fact that these sensors have non-linear response to the force as well as sensitivity differences between sensors. This enables one to obtain the curve that relates the value of the analogue output and the force.

However, more importantly, the proportionate contribution of each sensor is indicative of the manner in which load distribution takes place within the foot. An increase in the metatarsal sensors along with a decrease in the heel sensor would mean that there is redistribution of pressure to the forefoot. The manner of distribution of load is particularly significant when it comes to pointe dancing since the dancer is moving from one point of contact to another on the floor and pointe shoe.

Since the primary aim of this prototype is the detection of changes in the load distribution pattern and not the replacement of laboratory-quality force plates, the FSR-based technique acts as a cheap relative load measuring system.

4.2.2 Inertial Motion Measurement

MPU6050 provides complementary information on the movement of the foot that cannot be collected using pressure measurements alone. The IMU consists of a three-axis accelerometer and a three-axis gyroscope, which measure readings along the x, y, and z axes.

The accelerometer measures specific force and gives the information on linear acceleration and orientation of the gravitational field in relation to the sensor. The gyroscope measures angular velocity around all three axes. Angular velocity is of particular interest in performing actions like pirouettes, because it enables quantification of the rotational motion of the foot in terms of numbers and not merely visual estimation. Thus, the gyroscope is able to give the magnitude and progression of rotation of the movement.

However, integration of the measurements from the gyroscope over time, which would enable determining orientation, results in drift due to bias and noise. The accelerometer gives independent reference in the form of direction of the gravitational field. However, the readings from the accelerometer become inaccurate during dynamic movement, because they consist of gravitational and acceleration due to motion. Thus, the two measurements are fused together.

4.2.3 Madgwick Sensor Fusion

The Madgwick orientation filter is applied to integrate the measurements from the accelerometer and the gyroscope in order to provide an improved estimation of the foot orientation. Orientation is described using a quaternion, not the Euler angles.

The gyroscope is responsible for the short-time rotational dynamics of the foot, whereas the accelerometer measures the gravity field that could be used for correcting the errors in orientation that arise due to the integration process. Therefore, the orientation is continuously estimated using the data from the gyroscope and the accelerometer. Further, the orientation is represented in terms of angular variables – pitch, roll and yaw.

For Equipointe, this approach to the processing of the IMU measurements is crucial because raw IMU data is not the end-product in biomechanics research. The IMU data is only the initial information, and further the transformation of the measurements to estimates of the foot orientation and rotation will be performed.

4.3 Data Acquisition and Communication

The ESP32 serves as an intermediary that takes sensor readings and transfers them to the analysis software. While running, the microcontroller periodically takes readings from all four channels of FSRs while concurrently interacting with the MPU6050 through the I²C protocol. Each of the measurements gets timestamped to preserve the temporal relation of pressure and motion readings.

The readings are bundled in a structured data stream and sent via wireless connection to the dashboard implemented in Python through the WebSocket protocol. WebSocket was chosen for this purpose because it provides a constant two-way interaction between the microcontroller and computer without making an additional request for each measurement. This solution is suitable for monitoring of biomechanics since the signal readings have a built-in temporal nature and should be constantly received.

The dashboard has two purposes. Firstly, it provides real-time visualization of the sensor readings, thus allowing to monitor the performance of the hardware. Secondly, it stores the received data in CSV format, which allows for the retrospective analysis of the same measurements.

4.4 Experimental Testing and Prototype Validation

The proof of concept was analyzed using the controlled motion tests to see if the sensing device would be able to detect changes in loading and rotational motions that occur in ballet motion. Two main functions of the prototype were to be tested in the following tests – the ability of the FSR to detect changes in loading and the ability of the IMU to detect the change in rotation.

Static and controlled loading tests were carried out initially. Loading was applied to the pointe shoes in different configurations while at the same time the outputs from the individual FSR sensors were recorded. This was done to ensure that the sensors reacted to changes in loads as expected and also that there was a difference in signals at the four points of the sensors.

Dynamic testing was then carried out during ballet movements where the sensors' outputs were noted when there were changes in loading and rotation.

One benefit of gathering pressure and inertial data at the same time is that the two pieces of data can be analyzed together. This is because a time of higher rotational speed can be contrasted with the plantar loading data for the same time period. This means that Equipointe can progress from just identifying when movements have been made to analyzing the relationship between movements and foot loads.

4.5 Signal Processing and Feature Extraction

Sensor measurements were first post-processed before using them for biomechanical analysis. It was done due to the presence of noise, offsets and artefacts present in the sensor raw output data which could distort the real movements.

Pressure signals have been analyzed for the presence of baseline drift and noise components. If required, filter can be applied to minimize high-frequency signal components without altering the temporal dynamics of loading events. Similarly, IMU signals also require processing since gyroscope signals carry the sensor bias information and accelerometer signals are subjected to dynamic acceleration.

After sensor fusion, quaternions represent the continuous estimate of orientation of foot during each movement. Thus, angular displacement can be determined relative to any reference orientation. On the other hand, angular velocity can be obtained directly from the gyroscope signal and analyzed in combination with orientation information.

Thus, for each experiment, features from both sensors can be extracted. Features from pressure sensors include peak force, mean force, loading time and distribution of forces among different sensor positions. Features from the motion sensor include angular velocity, angular displacement and orientation changes.

The extraction of such variables converts the prototype to a biomechanical measuring system. Rather than analyzing thousands of individual measurements made by the sensors, it is now possible to look at the actual characteristics of the movement being analyzed and compare them across various trials.

4.6 Application to Ballet Movement Analysis

The final part of the proof of concept outlines how Equipointe can be used in ballet technique analysis. In ballet, the control of balance, rotation and plantar loading needs to be coordinated, so one measurement cannot describe the process properly. Therefore, FSR and IMU data complement each other in providing the necessary information.

For instance, during a pirouette, the IMU can measure the angular velocity and the orientation of the foot, whereas the FSRs will provide the information on how the load is distributed between different parts of the shoe. Thus, the two signals will be synchronized to determine correlations between the rotational activity and loading.

Thus, the prototype creates an initial base for the study of whether there is a connection between the parameters of the movement and stability. Particularly, with the help of the device, the values which cannot be measured objectively and need visual analysis can be quantified.

Equipointe's proof of concept shows the viability of embedding cheap pressure and inertial sensors into a pointe shoe and sending this data in real-time. While the prototype cannot provide as accurate readings of forces as laboratory force plates or spatial resolution as specialized motion tracking systems, the main value of this prototype is its ability to be portable. This allows dancers to gather biomechanical data during actual ballet steps and not inside a laboratory.

The built system, therefore, offers three related streams of information: plantar loads, inertial motion and time-accurate motion data. Combined together, this constitutes the technical basis for the future development of Equipointe as a wearable system for studying ballet biomechanics.

4.7 Corrective Feedback System

An important aspect of Equipointe is the transformation of biomechanical parameters into a corrective feedback for the dancer. Instead of just displaying the raw outputs of the sensors, this system is expected to detect movement parameters that are different from the pre-defined parameters of the movement and provide the dancer with the necessary corrective tool.

After the data collection and processing stage, certain biomechanical parameters are matched with the movement pattern references. Such parameters may vary and may be related to the plantar load distribution, angular velocity, foot orientation, and angular deviation. Once a discrepancy is detected, the error is categorized based on the movement parameter involved. For instance, excessive loading on the lateral or medial side of the foot can imply an improper weight distribution, while the excess of angular deviation can mean losing the alignment during the rotation movement.

Once the error is detected, it is matched to one of the pre-recorded corrective videos, each showing one particular technique mistake and the proper correction for it.

The above-mentioned approach will enable the system to give personalised feedback instead of the generic material being provided to all the dancers. In other words, if a dancer's data show a lot of deviations from plantar loading, he or she will be given information about the technique of

putting one's feet right and distributing weight; at the same time, a dancer showing instability in rotational orientation will receive information about how to align.

Thus, the above-mentioned corrective video system creates a feedback loop when the dancer's performance is constantly monitored and the obtained data is utilised during further practice sessions. In ballet, it is especially important since technical mistakes may be subtle, and they may not be recognised by dancers. The connection between quantified measures and specific teaching information shows how wearable sensors can be turned into an easily usable training tool instead of being only a measurement device.

Proof of concept shows the possibility of establishing the connection between the sensor-derived biomechanics and predefined technical error. In the future, the system can incorporate more sophisticated algorithms which will be able to analyse several features of sensors at once.

5.Dance Tech in India's Dancing Market

The Indian dance training market is an area of business that holds great potential and is growing rapidly. The current section provides relevant information on the size and growth of the market, its segmentation, and regional dynamics in relation to India, thus serving as a strong base for analyzing the commercial potential of the product under consideration.

5.1 Market Size and Growth Trajectory

The market for performance arts in India covering dance, music, and theater was estimated at USD 3.8 billion in 2022, where the addressable market is about 26 million users between ages 8-25 years, accounting for 7% penetration. Of these segments, the dancing training segment alone is a huge market. The 6-14 year age group is a USD 2 billion market in 2022 and is forecasted to become a USD 3.5 billion market by 2027. The overall market for dance schools in India, both online and offline, was estimated to be about INR 1,200-1,500 crore (USD 150-185 million) in 2019 before witnessing rapid growth prior to contracting due to the impact of the COVID-19 pandemic in 2020.

The online dance training sub-segment, while smaller, is growing at an accelerated pace. India is identified as a key growth engine within the **Asia-Pacific region, which accounts for 44% of new user sign-ups globally for online dance platforms**. The Asia-Pacific online dance training market is projected to grow from USD 2.48 billion in 2024 to USD 5.0 billion by 2035 at a CAGR of 6.6%, with India cited alongside China and Southeast Asia as a primary driver.

5.2 Market Segmentation

The dance market in India is segmented along multiple dimensions that reveal distinct user preferences and monetisation opportunities. These dimensions include dance style, age group and training format.

4.2.1 Segmentation by dance style

The market is wide and consists of genres like classical, folk, contemporary, and international genres. The classical genres like Bharatanatyam, Kathak, Odissi, and Kuchipudi still enjoy great cultural popularity, especially in South India and urban places. Bollywood dance styles and fusion styles appeal to the most youth population because of social media trends and films. Western dance forms, including Hip-Hop, Jazz, Contemporary, and Ballet, are becoming popular in cities, with Hip-Hop taking up 33 percent of genre preference among city youth. Folk regional dance forms like Bhangra, Garba, and Lavani are 31 percent more popular in their respective states.

5.2.2 Segmentation by age group

The target market comprises children and teens. The age bracket between 6 to 14 is made up of 15 million people who have the potential of being paying customers, and whose market share currently stands at \$2 billion. Another bracket, the one between the ages of 15 to 25, includes 11 million people who currently constitute a market worth \$1.8 billion. Within the dance school market, the young people below the age of 12 and the teenagers between 13 and 18 years form the major consumers of dance services, with the parents buying for the former and the teenagers themselves buying for the latter.

5.2.3 Segmentation by training format

The offline dance schools and dance studio continue to dominate the space, especially in Tier 1 and Tier 2 cities, but online learning is gaining ground at an unprecedented rate. Digital learning gained momentum during the pandemic period with many schools adopting hybrid methods of learning using both physical and digital classrooms. The digital medium provides flexibility and convenience for the learners, which suits people who live in far-off places and learners with erratic timings. The mobile-first distribution channel is dominant, with 68% of the learners using smartphones and 74% of the beginners enrolling via mobile applications.

5.3 Regional dynamics within India

The dance training market exhibits significant regional variation, reflecting cultural diversity, economic development, and urbanization patterns.

5.3.1 Tier 1 cities (Mumbai, Delhi, Bangalore, Chennai, Hyderabad, Kolkata)

These urban areas are responsible for most structured dance classes, having high disposable incomes, well-established dance schools and awareness of world trends. The central and northwestern parts of the market have shown the most active dynamics, having the highest prices of subscriptions and the highest density of premium schools.

5.3.2 Tier 2 and tier 3 cities

This is one of the most dynamically developing segments with 42% of growth coming from this group. It is supported by affordable prices, multilingual content and mobile-first strategy. Localized content in native language helps to increase engagement by 39%, regional dances have higher retention in their home state.

5.3.3 Rural and semi-urban areas

Pervasiveness is still relatively low because of poor internet connectivity, compatibility, and cultural constraints. However, government campaigns for digital literacy and increased availability of inexpensive mobile phones are gradually bridging the gap.

5.4 Key Market Players and competitive landscape

There are various online and traditional studios involved in the industry in India.

5.4.1 Offline Studios and Academies

Well-known players in the space include Terence Lewis Performing Arts, Shiamak Davar's Dance Company, Naadyog, Danceworx, and thousands of other independent studios located in various regions of the country. The industry is quite fragmented, where the top 5 participants control only a small part of the market.

5.4.2 Online Platforms

Some of the global platforms operating in India include Steezy, CLI Studios, and Udemy, while some local platforms include Choreo N Concept, Groove Therapy, and Neetu's Dance & Fitness. Terence Lewis, one of the popular Indian choreographers, has his digital platform which takes advantage of his brand name. These platforms provide a combination of live and recorded videos at subscription fees between INR 500 and INR 2,000 per month.

5.4.3 Hybrid Models

Most of the traditional dance studios have now opted for a hybrid model, providing both online and in-person classes post-pandemic to keep their students and attract more customers. This model is preferred by those who seek flexibility but still want some face-to-face interactions and feedback.

5.5 Investment Trends and Opportunities

5.5.1 Funding and acquisition

Although there are no India-specific deals available, the general trend suggests high investment activity. In 2023, TZP Group acquired Break The Floor Productions and Star Dance Alliance and created DanceOne Holdings to create an entity in the U.S., a similar model which can be followed in India as well. DNCR raised \$15 million for platform improvements.

5.5.2 Growth opportunities

The second-tier and third-tier cities present the most significant growth opportunity, contributing to 42% of the new user additions, while having no physical studio presence. Corporate wellness partnerships, K-12 dance programs, and micro-credentials certifications provide other sources of revenue. Region-specific content and fusion formats combining the old and new styles are other ways to attract customers.

5.5.3 Technology integration

AI-powered feedback, gamification, and AR/VR experiences are emerging differentiators. Platforms that integrate motion tracking, real-time corrections, and personalized learning paths can command premium pricing and improve retention.

5.6 Challenges and Limitations

In spite of all this growth, there are indeed a number of challenges facing the dance market in India. This is a very sensitive market in terms of economic changes and fluctuations because the demand is linked to disposable income levels; in bad times, one of the first things to be cut back is training in dance, as was seen in 2020 where the demand fell sharply due to the impact of the pandemic. There are high costs associated with this industry, with studio membership costs being between INR 2,000 and INR 8,000 on average, while the cost for online memberships is between INR 500 and INR 2,000, which limits access for poorer sections in rural and semi-urban parts. There is a lack of availability of certified instructors, who mainly work in Tier 1 cities. In terms of digital infrastructure, there are several gaps that prevent people from using the online services: 36% mention device issues (old smartphone, insufficient space), whereas 29% state that their problem is bandwidth. Engaging users and making them stick to the platform is a common problem: 39% of beginners leave the website within several lessons because they miss a physical presence, direct feedback and accountability, which an in-person lesson provides naturally. Culture is another barrier, as older people (who are aged over 45) find it inappropriate to engage in dancing due to existing stereotypes about what activities are suitable at a certain age. In some countries and regions, gender norms prevent women from engaging in some types of dance, and vice versa. Finally, there are certain challenges related to unit economics: companies have to provide discounts to gain new users, while maintaining quality of instructions, protecting minors, dealing with music licensing, and building a viable business model.

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Final Paper Outline

1. Introduction

2. Research Objectives and Methodology - 1.5 pages

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- Analyse various types of risks
-

b. Research Question

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i.

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2.2 Costs and risks associated with training

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 - Decreased endurance and movement efficiency
 - Missed rehearsals and performances
 - Interrupted skill acquisition and delayed career progression
- Economic costs
 - Direct medical rehabilitation costs
 - Indirect costs: Equipment and training modifications
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- Limitations of current prevention
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4.3 Regional dynamics within India

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- Offline studios and academies
- Online platforms
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- Challenges and limitations

6. Findings from Primary Data - 2 Pages

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Biomechanics & Injury

Biomechanics & Injury – 1000 words

- <https://www.benjamindombmd.com/blog/what-are-common-injuries-for-ballet-dancers/?bp=40201>
- What is the science of biomechanics behind Ballet ?
 - Ballet biomechanics involve extreme precision, manipulation of human anatomy, utilising external hip rotation, vertical alignment and high velocity movements that often produce ground reaction forces 4-5 times a dancers' body weight. It centres around optimising energy transfer and stability through controlled muscle torque and various rotational mechanisms.
- List down various types of injuries and risks that ballet dancers are exposed to.
 - Achilles Tendonitis (ankle): Ballet causes achilles tendonitis due to repetitive, high intensity strain on the tendon, biomechanically known as 'overuse.' It may also be caused due to excessive **plantarflexion**². Early symptoms include heel/ankle pain, stiffness and swelling.
 - Ankle Sprains: This is caused due to the errors in technique 'sickling' and 'rolling.' Early symptoms of this include sudden pain, swelling and reduced mobility.
 - Stress Fractures: stress fractures are caused due to poor weight distribution on the sole of the foot. Early symptoms of stress fractures include: persistent localised pain and swelling.
 - Plantar Fasciitis: This is caused due to repetitive jumping movements (known as allégro in ballet), repetitive relevés (rising) and working in pointe shoes which leads to micro tears in the fascia. These injuries are augmented by dancing on hard, non conducive surfaces and maintaining a constant, high tension stretch in the plantar fascia during intense training.
 - Patellofemoral Pain: Ballet causes patellofemoral pain primarily through repetitive overloading of the knee joint, often stemming from poor alignment, weak glute muscles and forced turnout.

² Plantarflexion is the movement of the ankle where the foot points downward, away from the shin, increasing the angle between the foot and leg.

- Spondylolysis (lower back): Spondylolysis is a stress fracture in the lower back caused primarily through repetitive hyperextension (arching) of the lumbar region.
- Summarise:
 - Li F, Adrien N, He Y. Biomechanical Risks Associated with Foot and Ankle Injuries in Ballet Dancers: A Systematic Review. Int J Environ Res Public Health. 2022 Apr 18;19(8):4916. doi: 10.3390/ijerph19084916. PMID: 35457783; PMCID: PMC9029463.

Ballet biomechanics involve extreme precision, manipulation of human anatomy, utilising external hip rotation, vertical alignment and high velocity movements that often produce ground reaction forces 4-5 times a dancers' body weight. It centres around optimising energy transfer and stability through controlled muscle torque and various rotational mechanisms.



Ballet biomechanics runs on two primary principals: Newton's laws of motion in ballet and balance and stability. This reflects the fact that movements in dancing are deeply scientific, rather than arbitrary. Newton's first law related to inertia can be seen when a dancer rotates in a pirouette, for example, as they continue rotating until obstructed by an opposing force, like

friction or muscle breaking. Newton's second law, $F=ma$ is seen through jumps, where jump height is dependent on the force exerted on the floor relative to body mass. Lastly, Newton's third law on action and reaction forces is reflected through jumps as well as the downward push of the foot on the floor generates an equal and opposite force that propels the dancer into a jump. Furthermore, stability and balance in ballet depend on three key factors. First, the centre of gravity. Typically located near the pelvis, the centre of gravity must be aligned with the supporting leg to ensure balance while dancing. Second, the base of support (the dancer's feet or single foot) requires neuromuscular control. Third, alignment plays an important role as the shoulders, knees and hips need to be 'stacked' on top of each other to minimise compensatory limb activity and maximise efficiency.

Research identifies four biomechanically grounded pillars of classical ballet technique: turnout, the external rotation of the hips; alignment, the vertical stacking of joints to minimise shear forces; placement, the optimal positioning of body segments relative to the centre of gravity; and extension, the controlled elongation of the limbs while maintaining alignment.

In ballet, lower-body muscle activation and joint kinetics revolve around the coordinated action of the quadriceps, hamstrings, glutes, calves, hip flexors, and deep external rotators to produce and control movement at the hip, knee, and ankle, while the core and spinal muscles stabilize the trunk and transmit forces efficiently. The quadriceps extend the knee during jumps and control landing, hamstrings and glutes drive hip extension in movements like arabesque and help stabilize the pelvis, and the calf complex generates plantarflexion for relevé and pointe work while absorbing impact on landing. Hip flexors elevate the leg in développés and grands battements, and the deep hip external rotators enable turnout without dangerous compensations at the knee and ankle; simultaneously, the abdominals, erector spinae, and multifidi maintain spinal alignment and allow controlled extension in cambré and penché. In the upper body, scapular stabilizers and shoulder musculature support port de bras and partnering, integrating with lower-limb kinetics so that ground reaction forces, joint torques, and muscle activation patterns combine to create both the aesthetic lines and the mechanical efficiency characteristic of classical ballet technique.

Due to the physically demanding nature of ballet as an activity, dancers are prone to numerous injuries. Ballet causes achilles tendonitis due to repetitive, high intensity strain on the tendon, biomechanically known as 'overuse.' It may also be caused due to excessive plantarflexion³. Early symptoms include heel/ankle pain, stiffness and swelling. Ankle sprains may also be caused due to the errors in technique 'sickling' and 'rolling.' Sickling is when the alignment of the foot gets disrupted, while rolling is when the weight is distributed unevenly on the sole of the foot. Early symptoms of this include sudden pain, swelling and reduced mobility. Stress fractures are another such injury caused due to poor weight distribution on the sole of the foot. Early symptoms of stress fractures include persistent localised pain and swelling. Plantar Fasciitis is caused due to repetitive jumping movements (known as allégro in ballet), repetitive relevés (rising) and working in pointe shoes which leads to micro tears in the fascia. These

3 Plantarflexion is the movement of the ankle where the foot points downward, away from the shin, increasing the angle between the foot and leg.

injuries are augmented by dancing on hard, non conducive surfaces and maintaining a constant, high tension stretch in the plantar fascia during intense training. Ballet causes patellofemoral pain primarily through repetitive overloading of the knee joint, often stemming from poor alignment, weak glute muscles and forced turnout. Lastly, spondylolysis is a stress fracture in the lower back caused primarily through repetitive hyperextension (arching) of the lumbar region caused by certain movements in ballet.

Cost of Learning & Performing Ballet

Cost of Learning & Performing Ballet

India:

Average cost of lessons per year = Rs. 1,00,000

Average cost of equipment for one year (uniform, extra private rehearsals etc) = Rs. 50,000

Europe: France, Russia, Germany:

Dance is highly subsidised in these countries. Hence, with strong state-funded ballet institutions, elite training can be nearly free of cost in these countries. However, in other countries like the US, ballet might be more expensive because these systems are not in place.

Government Support for Ballet/ Performative Arts/ Dance

USA:

\$1500 per year for beginner recreational classes

\$20,000 - \$30,000 per year for pre-professional or boarding school training

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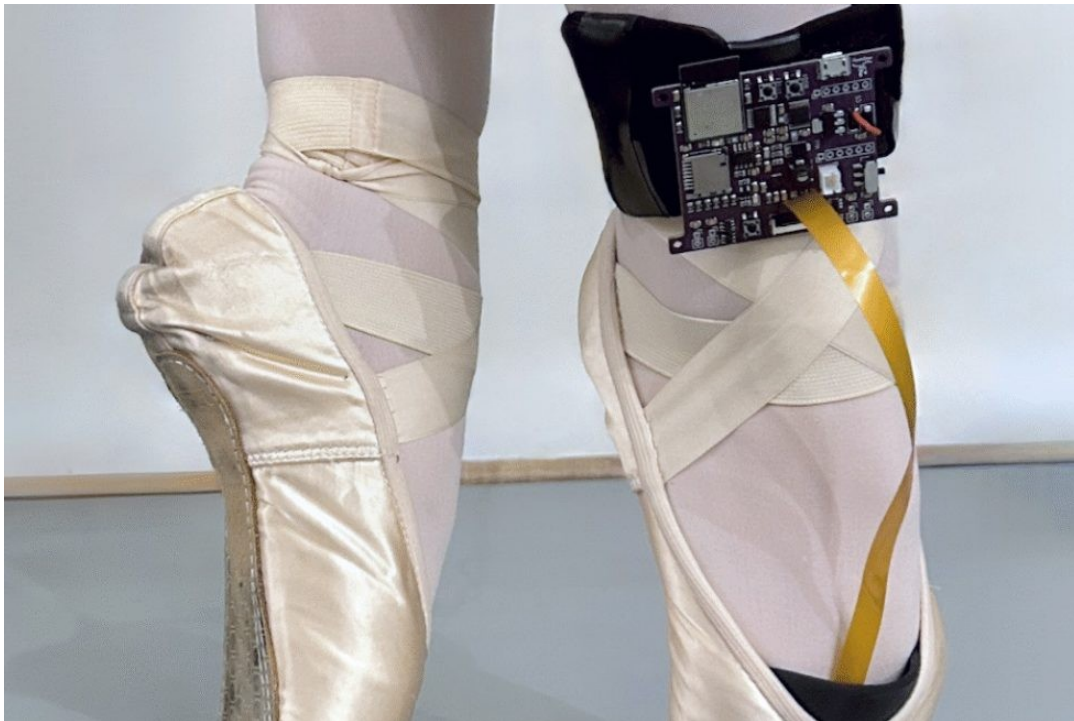
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Wearable Technology

Para 2: Thematic cluster 2 - wearable technology

In this section, we provide a comprehensive review of existing literature on wearable technology relevant to designing ballet shoes for injury prevention. A study conducted by scholars⁴ on creating a wearable device for ballet dancers to analyse pressure distribution using 64 pressure sensors in the front part of a ballet dancer's foot which sends signals to the dancer through an ankle band that provides feedback via haptics was conducted. The sensors are inserted into the shoe in the form of a toe pad and the sensors are flexible to accommodate this. The findings after studying this were that dancers with diverse databases with different foot flexibility, strength and shapes are needed to test this product; generalising the product is difficult as dancers have personalised needs; and making the system untethered, lightweight and unobtrusive is challenging due to space constraints in the pointe shoe and maintaining the mobility of the dancer.



Citation: <https://engineering.jhu.edu/news/engineering-meets-ballet-a-smart-solution-for-safer-pointe-work/>

⁴ “Engineering Meets Ballet: A Smart Solution for Safer Pointe Work”

This paper⁵ studied a between new pointe shoes that had 5-50 hours of training and dead pointe shoes that had 51-400 hours. The movements tested on dancers were relevé à la second, sous-sus and a single pirouette. The measuring the centre of pressure sway area using a force plate and Ground Reaction Forces (GRF) in three directions: anterior-posterior, medial lateral and vertical the following was found: The centre of pressure sway area, using a force plate and Ground Reaction Forces (GRF) was measured in three directions: anterior-posterior, medial lateral and vertical. The findings were that “Dead” shoes showed a significantly larger ‘sway area’ from the mean centre of pressure, impacting stability; No statistically significant differences in the force directions, vertical, medial and lateral, were found across the two shoe conditions; and “dead” shoes provided less stability and reduced balance, whereas ‘new’ shoes allowed more force production in certain movements.

- “Dead” shoes showed a significantly larger ‘sway area’ from the mean centre of pressure, impacting stability
- No statistically significant differences in force directions (vertical, medial lateral) across the two shoe conditions
- ‘Dead’ shoes provided less stability and reduced balance whereas ‘new’ shoes allowed more force production in certain movements

Citation: <https://thesportjournal.org/article/biomechanical-comparison-of-dead-and-new-pointe-shoes-in-female-professional-ballet-dancers/>

1. “A Wearable Insole System to Measure Plantar Pressure and Shear Stress” -

A study conducted on testing a wearable insole system to measure plantar pressure and shear stress for diabetic foot ulcer prevention details the design, development and testing of this system. The system utilises thin, flexible TRIPS sensors placed at the heel, 1st and 5th metatarsals and the hallux. The sensors recorded pressure and shear in two orthogonal directions and are integrated into an insole. The final product showed high accuracy demonstrating that the multi-sensor approach and flexible insole design is feasible for monitoring pressure distribution in pointe shoes.

Citation: <https://doi.org/10.3390/s23063126>

5 “Biomechanical Comparison of ‘Dead’ and ‘New’ Pointe Shoes in Female Professional Ballet Dancers (2019)”

2. **“The Pressure Pointe (2025)”** - this project aimed to reduce high pressure on a dancers' foot in pointe shoes, by specifically improving upon the pressure distribution in the toe box of the pointe shoes. The project focused on different materials inserted into the box to find which one was most effective in redistributing pressure among the toes, especially reducing stress on the big toe. A model foot was used to test the product, rather than a human. Several prototype inserts were created made of materials like spray foam, soft foam, rigid plastic, memory foam etc. They used pressure indicating film to identify the pressure distribution. Through conducting this experiment, the following was found:

- Among all materials tested, spray foam performed the best as it had the lowest average percentage area of toe that was unloaded. This means most of the pressure was distributed equally amongst all five toes.
- The second best was soft foam and the worst material was rigid plastic
- Spray foam was also liked by consumers as they found it improves support and comfort

A project aiming to reduce high pressure on a dancers' foot in pointe shoes, by specifically improving upon the pressure distribution in the toe box of the pointe shoe focused on different materials inserted into the box, to investigate which was most effective in redistributing pressure amongst the toes with a specific focus on reducing pressure on the big toe. The materials tested include spray foam, soft foam, rigid plastic, memory foam and more. Pressure indicating film was used to identify the pressure distribution and it was concluded from the experiment that amongst all materials tested, spray foam performed the best as it had the lowest average percentage area of the toe that was unloaded; the second best was soft foam and the worst material was rigid plastic; and spray foam was also liked by consumers as they found it improved support and comfort.

Citation: <https://www.sensorprod.com/research-articles/the-pressure-pointe/>

3. **“Sensing Technology for Assessing Motor Behaviour in Ballet (2022)”** - This paper provides a comprehensive summary of sensing technologies used to assess motor behaviour in ballet. After examining various studies, it was concluded that force plates, electromyography (EMG) and inertial sensors are the most frequently used methods for evaluating classical ballet movements. Certain important findings from this paper are:

- 55% of studies used two or more types of technology, most commonly combining kinematic (motion capture) and kinetic (force plate) analyses for better data

- Only 29 distinct ballet movements have been analysed and studied in detail with sauté, grande jeté and pas de chat being studied the most
- EMG is used to analyse muscle activation in movements like pliés

A study was conducted on the most effective sensing technologies used to assess motor behaviour on ballet. After examining various studies, it was concluded that force plates, electromyography (EMG) and inertial sensors are the most frequently used methods for evaluating classical ballet movements.

Citation: <https://doi.org/10.1186/s40798-022-00429-8>.

4. **“A Smart Wearable Device for Monitoring and Self-Assessment in Sports”** - This study presents a smart wearable device for sports and rehabilitation applications that can monitor and self-assess movement using multi-sensor technology. It integrates pressure sensors, temperature sensors, and moisture sensors within a shoe to continuously collect and wirelessly transmit data. These sensors accurately map plantar pressure and environmental parameters during activity and send data wirelessly to a paired app or receiver for analysis and feedback. Results show it reliably tracks weight distribution, detects changes in gait and loading, and contributes valuable feedback for athlete training, rehabilitation, and injury prevention.

Citation: <https://doi.org/10.1016/j.ijmedinf.2020.104343>

5. **“Pressure Measurement on Toes in Pointe Shoes”** - This study introduces a novel pressure measurement system using thin, flexible capacitive sensors embedded in pointe shoes to quantify toe pressure distribution during ballet movements. The sensors, placed across the toe box, captured real-time data on pressure hotspots in professional dancers performing relevés and jumps, revealing peak loads exceeding 200% body weight on the first metatarsophalangeal joint. Key findings include significant pressure asymmetry between toes, with the hallux bearing 40-50% of total load, and correlations between sensor data and self-reported foot pain.

Challenges addressed in the study:

- Sensor flexibility to avoid altering shoe dynamics
- Wireless data transmission for untethered performance
- Calibration for varying foot shapes

A study was conducted on a novel pressure measurement system, leveraging thin, flexible capacitive sensors embedded in pointe shoes to quantify toe pressure distribution during ballet movements. The sensors, pulled across the toe box, captured real-time data on pressure hotspots in professional dancers performing relevés and jumps, revealing peak loads exceeding 200% body weight on the first metatarsophalangeal joint. Key findings include significant pressure asymmetry between toes, with the hallux bearing 40-50% of the total load, and correlations between sensor data and self-reported foot pain.

Citation: <https://journals.sagepub.com/doi/10.1177/1089313X251330462>

6. **“Wearable Accelerometer for Ballet Jump Analysis”** - This paper evaluates a shoe-mounted accelerometer and gyroscope system for assessing ballet jump height, landing forces, and aerial time in variations like sautés and jetés. Worn by 12 dancers, the IMU sensors provided 95% accuracy against force plates, identifying technique flaws linked to sesamoiditis from harsh landings. The untethered setup enabled field testing without lab constraints.

Key findings:

- Jump heights averaged 0.4m with 2-3x body weight impacts
- Real time feedback improved landing symmetry

A paper evaluating a shoe-mounted accelerometer and gyroscope system for assessing ballet jump height, landing forces, and aerial time in variations like sautés and jetés was worn by 12 dancers. The IMU sensors provided 95% accuracy against force plates, identifying technique flaws linked to sesamoiditis from harsh landings. The untethered setup enabled field testing without lab constraints. Another key finding was that real time feedback improved landing symmetry for jumps.

Citation: <https://pubmed.ncbi.nlm.nih.gov/31718476/>

7. **“Pressure Distribution and Foot Injury in Ballet Dancers”** - The research used force-sensitive insoles in pointe shoes to analyze pressure patterns across 20 professional ballerinas during pliés, relevés, and pointe work. Results showed metatarsal heads endured 1.5-2.5x normal loads, with rigid shanks exacerbating hallux rigidus risks; flexible toe pads mitigated peaks by 25%

Relevant challenges:

- Individual foot morphologies required custom sensor arrays

- Long term wear caused sensor drift (this is something to take into consideration while building our prototype)

Citation: <https://pubmed.ncbi.nlm.nih.gov/22687718/>

8. **“Measuring Foot Pressure in Pointe Shoes of Ballet Dancers”** - Researchers developed a wearable insole with 16 FSR sensors integrated into pointe shoes to map plantar pressure during dynamic ballet tasks like pirouettes and grand battements. The system logged data at 100 Hz, showing "dead" shoes increased peak pressures by 15-20% under the metatarsals compared to new ones, heightening injury risk. Haptic feedback prototypes reduced overload incidents by 30% in trials with elite dancers.

Notable outcomes:

- Higher sway in centre of pressure with worn shoes
- Need for dancer specific calibration
- Lightweight design under 20g per shoe

Citation: <https://pubmed.ncbi.nlm.nih.gov/40098421/>

Reference

Academy, U. S. (2019, February 13). *Biomechanical comparison of “Dead” and “New” pointe shoes in female professional ballet dancers*. The Sport Journal.

<https://thesportjournal.org/article/biomechanical-comparison-of-dead-and-new-pointe-shoes-in-female-professional-ballet-dancers/>

Engineering Meets Ballet: A Smart Solution for safer pointe work - Johns Hopkins

Whiting School of Engineering. (2025, July 1). Johns Hopkins Whiting School of Engineering. <https://engineering.jhu.edu/news/engineering-meets-ballet-a-smart-solution-for-safer-pointe-work/>

intro + lit review

Dance technology has been an area of inquiry for both business and engineering scholars (see: Judith A. Gray 1989).

Tab 6

Skeleton

1. Introduction

a. Introduction is also providing some historical background to this dance form

2. Biomechanics and injury in Ballet Training

a. Common Injuries in Ballet Training

b. Costs and Risks associated with training

3. Wearable tech

4. Proof of Concept of the Sensor

5. Market Analysis

6. Conclusion